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Paper title: A K-Means–based Gaussian Process Regression Model for Human Motion Prediction in Human–Robot Collaboration

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ONE LOVE. ONE FUTURE.

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I. Introduction

1. Background and Motivation

- Human Robot Collaboration (HRC) requires robots to anticipate human motion to avoid collisions and improve task efficiency.
- Human motion is nonlinear and stochastic, making accurate prediction challenging.
- Predictions without confidence estimates may cause overconfident robot behavior, uncertainty quantification is essential for safety.

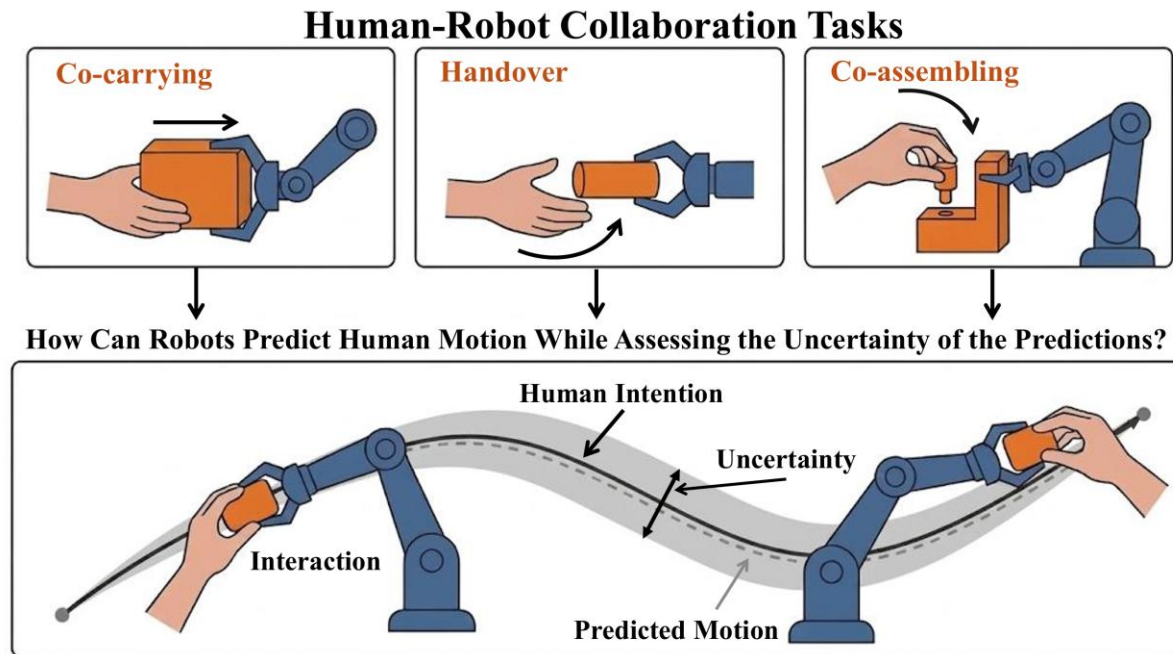


Fig. 1: Illustration of common human-robot collaboration scenarios

I. Introduction

2. Research Gap

- Model-based methods rely on explicit physical models, making them struggle with nonlinear and stochastic human motion.
- RNN / LSTM / GRU can capture temporal dependencies but require large datasets and lack uncertainty quantification.
- Standard Gaussian Process Regression (GPR) provides uncertainty estimates but suffers from $O(N^3)$ complexity → Infeasible for real-time applications.

3. Main Contributions

- A K-Mean-based Gaussian Process Regression (KMGPR) model that reduces computational cost while maintaining prediction accuracy comparable to standard GPR.
- Uncertainty quantification through predictive variance, validated on two representative HRC scenarios.
- Theoretical analysis explaining the prediction stability differences between KMGPR and standard GPR under abrupt motion changes.

II. Proposed Method

1. Multi-dimensional GP Architecture

Problem Formulation:

- Input: $\mathbf{X}_t = [\mathbf{p}_{t-h+1}^\top, \dots, \mathbf{p}_t^\top] \in \mathbb{R}^{dh}$
- Output: $\mathbf{Y}_t = \mathbf{p}_{t+1}^\top \in \mathbb{R}^d$

For each dimension:

- Predictive mean: $\mu_{y_*} = \mathbf{K}_*(\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}$.
- Predictive variance: $\Sigma_{y_*} = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{K}_*(\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{K}_*^T$

⇒ Requires $N \times N$ matrix inversion → $O(N^3)$ complexity

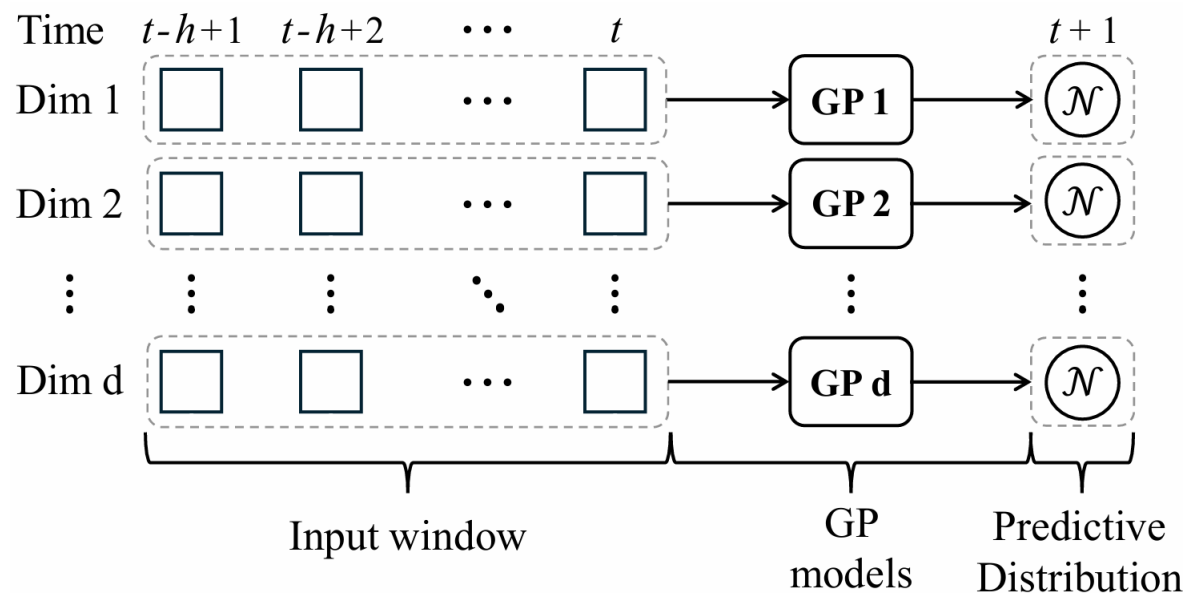


Fig. 2: Multi-dimensional GP architecture

II. Proposed Method

2. Proposed KMGPR Framework

- **Training phase:** The Elbow method-based K-Means selects a representative subset ($M \ll N$) from the full training dataset.
- **Prediction phase:** Replacing the full training dataset with the representative subset reduces the time complexity from $O(N^3)$ to $O(M^3)$.

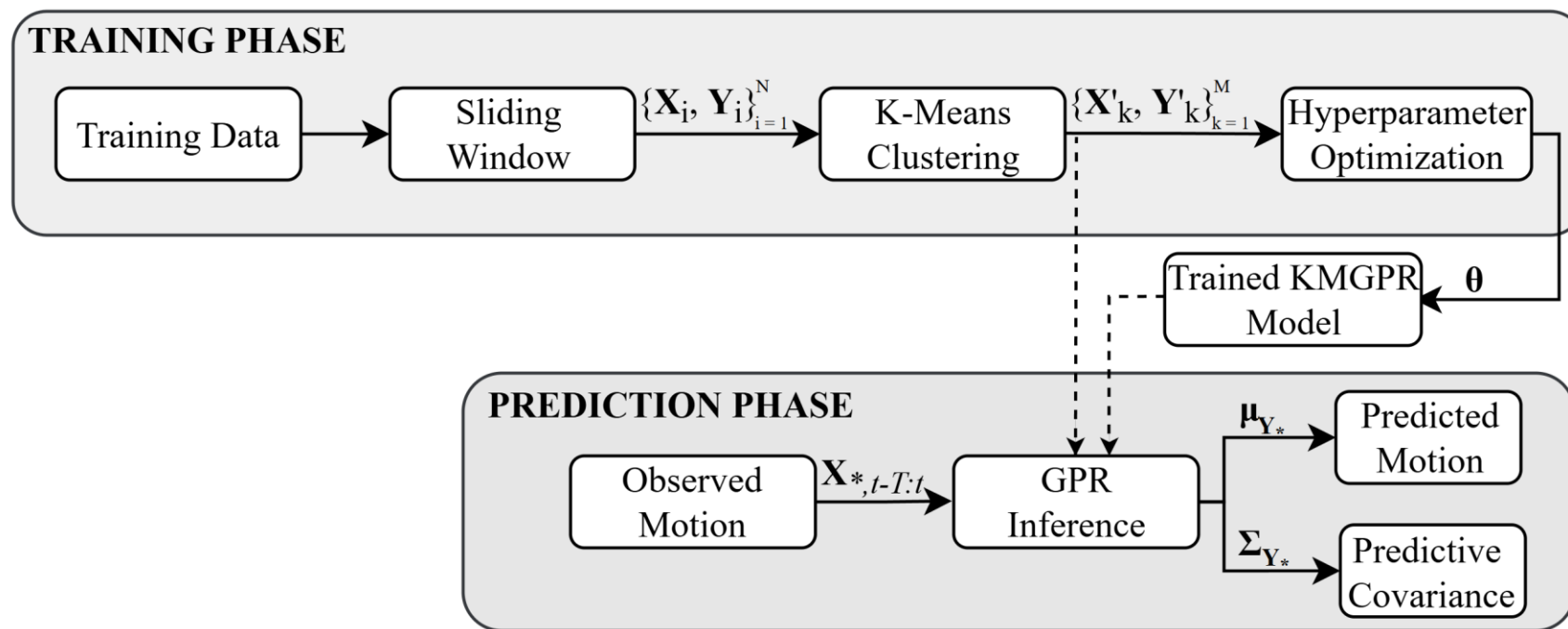


Fig. 3: The framework for KMGPR

III. Validation Example

1. Validation Setup

- 30 trajectories from initial position (0, 0) toward 3 distinct targets within 10s at a sampling interval of 0.1s
- 80% of trajectories used for training, generating 2424 sliding windows
- The Elbow method-based K-Means selects 484 representative windows the full dataset, retaining only 20% of the original windows

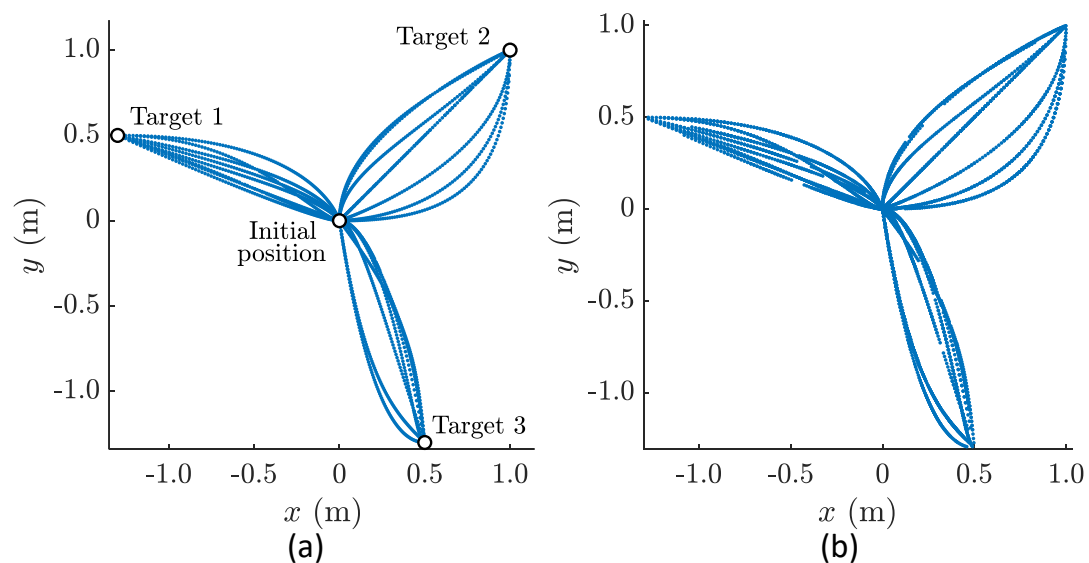


Fig 4: Distribution of training data: (a) the full dataset
(b) the subset of data using K-Means

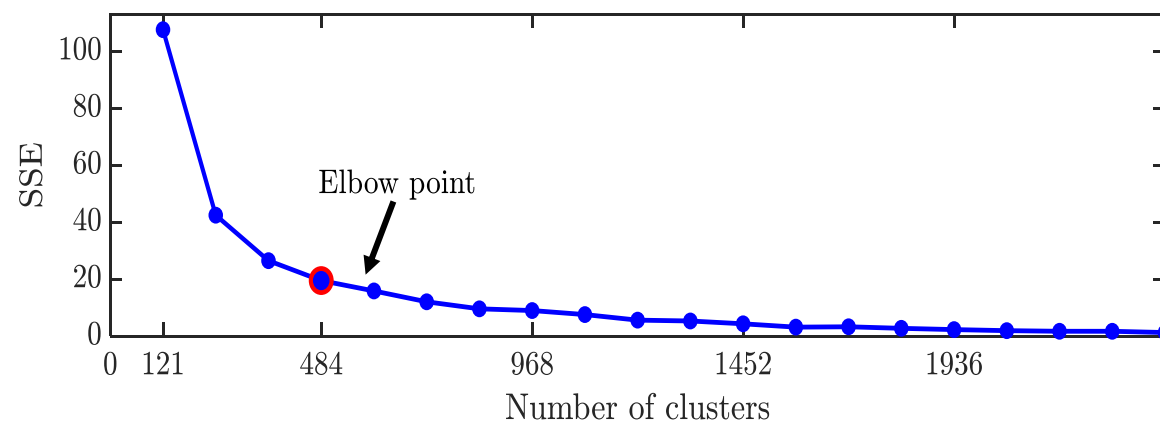


Fig. 5: The SSE curve using the Elbow method for determining the optimal number of clusters

III. Validation Example

1. Validation Setup

Table I. Parameters for model training

Hardware	Intel Core i5-12400, 16 GB RAM
Window size	10
Kernel function	SE-ARD
Number of iterations	250
Initial learning rate	0.05
Optimizer	Adam optimizer
Evaluation Metrics	MSE

Objective function: Log-marginal likelihood

$$\log p(\mathbf{y}) = \underbrace{-\frac{1}{2}\log|2\pi(\mathbf{K} + \sigma_n^2\mathbf{I})|}_{\text{Complexity penalty}} \underbrace{-\frac{1}{2}\mathbf{y}^T(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}\mathbf{y}}_{\text{Data-fit term}} - \frac{N}{2}\log 2\pi.$$

III. Validation Example

2. Validation Results and Analytics

Scenario 1: Nominal Trajectories

Table II

Performance comparison on nominal trajectories

Model	MSE ($\times 10^{-6}\text{m}^2$)				Prediction Time (s)
	Target 1	Target 2	Target 3	Overall	
KMGPR	0.12	0.40	0.12	0.21	4.62
GPR	0.09	0.36	0.10	0.19	41.09

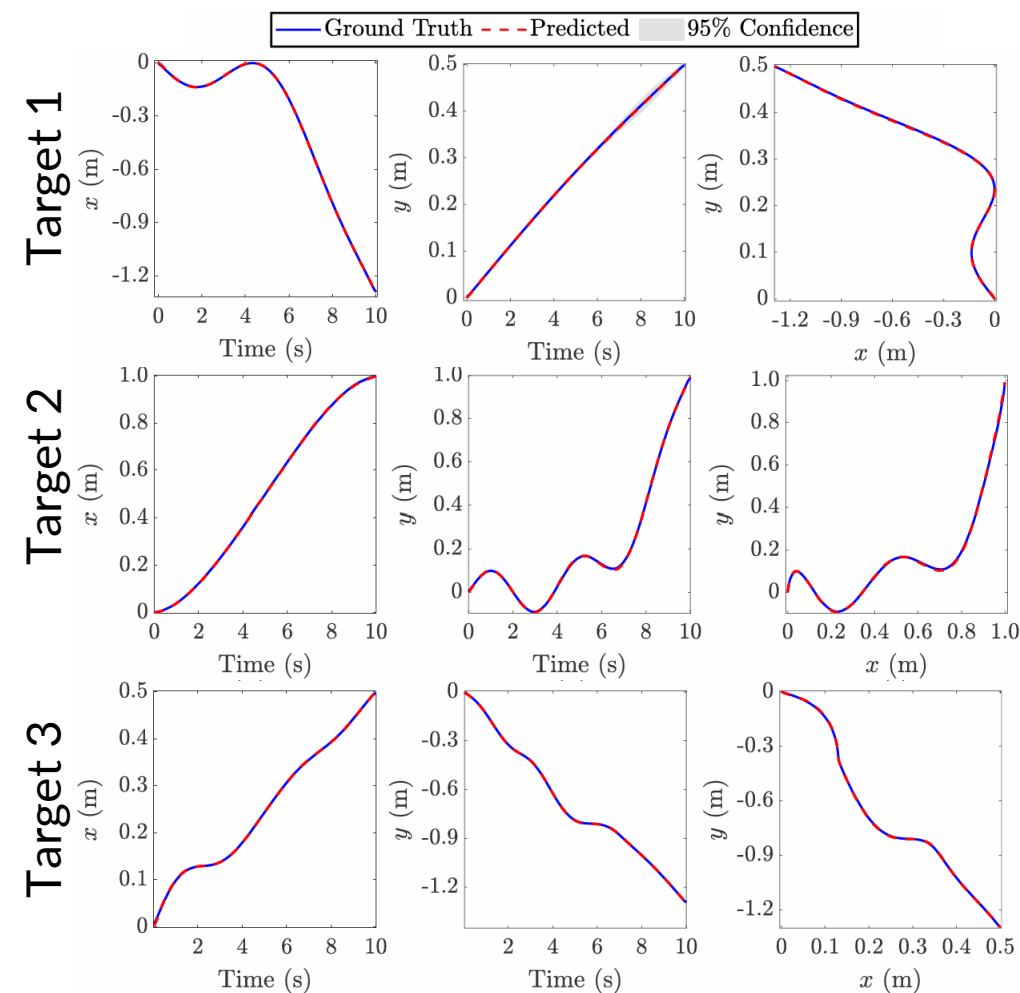


Fig. 6. Prediction performance of KMGPR with 95% confidence intervals on nominal trajectories

III. Validation Example

2. Validation Results and Analytics

Scenario 2: Abrupt Motion Changes

- KMGPR learns larger length scales, producing a smoother predictive function.
- GPR's smaller length scales cause sharp variations, leading to oscillatory predictions.

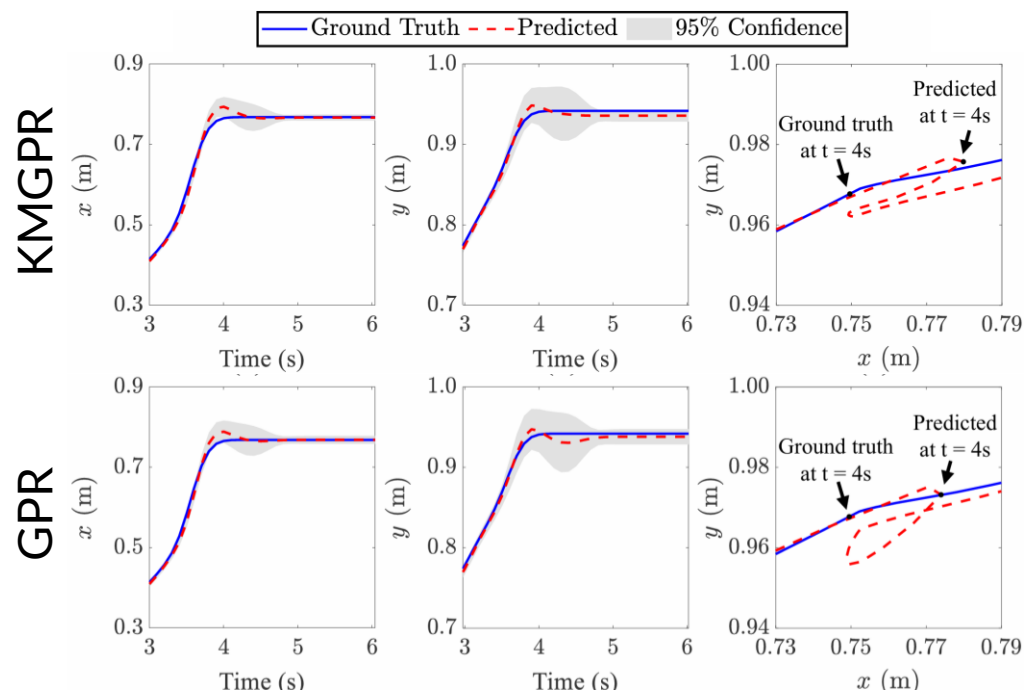


Fig. 6. Predictions under abrupt motion changes between KMGPR and standard GPR

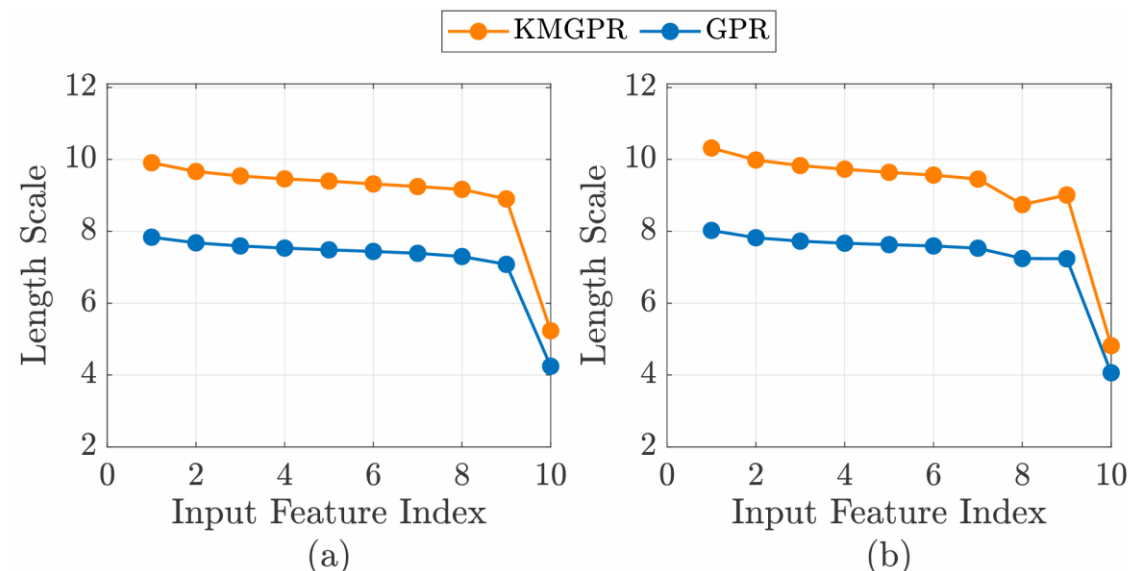


Fig. 7. Comparison of length scales across input features for GPR and KMGPR on (a) x-axis, (b) y-axis

IV. Conclusion

1. Summary

- KMGPR maintains prediction accuracy comparable to standard GPR while significantly reducing computational cost.
- The model effectively captures uncertainty through the predictive variance
- Under abrupt motion changes, KMGPR exhibits greater prediction stability with significantly fewer oscillations compared to standard GPR.

2. Limitations

- The model is validated only on two-dimensional simulated data generated in MATLAB.
- The subset selection via K-Means is performed offline and remains fixed after training.

3. Future Works

- Validate the model with real-world physical human-robot interaction in a three-dimensional framework.
- Develop adaptive control schemes that leverage predicted uncertainty for real-time decision-making.



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THANK YOU !